



GoalFlow

Features & Functionality

What the product does, and why each part exists

Stack Flutter · Node.js + TypeScript + Express · MongoDB Atlas · FCM · Email

Demo login demo@goalflow.app / Demo1234

A personal goal-tracking mobile app. Define what you want to achieve, break it into milestones and actions, and see honestly whether you are on track.

Flutter (Android + iOS) · **Node.js + TypeScript + Express** · **MongoDB Atlas** · **FCM** (push) · **Email** (OTP, reset, digests)

Demo login	demo@goalflow.app / Demo1234
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1. The product idea

Most goal apps are a to-do list with a due date. They tell you *what* you wrote down, never whether you are actually going to get there.

GoalFlow answers one question every time you open it: **"What should I focus on today, and am I on track?"** Everything in the product exists to answer that honestly — which means it will tell you when you are behind, and why, in a sentence you can act on.

The structure is three levels deep, because a goal you cannot start today is not a plan:

Goal	Learn Spanish	35% · On track
Milestone	Build basic vocabulary	60%
Action	Learn 20 new words	25 min · Mon–Fri · 07:30
Action	Complete one lesson	25 min · Mon–Fri · 07:30
Milestone	Improve conversation	0%
Action	Practise speaking aloud	
Action	Review past vocabulary	

2. Onboarding

Five short steps, then one atomic commit. A user can never end up half-onboarded with a profile but no goal.

Step	Collected
1	Name, optional profile photo
2	Main objective — the one thing this year is about
3	First goal: title, why it matters, category, target date
4	Preferred days and time of day
5	Session length and weekly target

Those answers are not stored and forgotten. They become:

- the default routine for every goal created later
- the default duration on every action
- the time reminders are scheduled for
- the tone of the home greeting — a morning person who opens the app at 4pm is told *"Your usual morning slot has passed — a short version still counts"*

Each new goal also arrives pre-filled with a **starter breakdown** for its category (Health, Learning, Career, Finance, Personal, Relationships, Productivity, Custom), so the user edits a plan rather than inventing one from a blank screen.

3. Goal management

Create, edit, **pause**, **resume**, **complete** and archive.

Field	Notes
Title, description	
Why it matters	Motivation is part of the product, not decoration
Category	8 built-in + custom
Priority	Low / Medium / High
Start and target date	
Routine	Daily, specific days, N times per week, or one-off
Preferred time, duration	Inherited from onboarding, editable per goal
Colour	Used across cards, calendar and charts

Two behaviours worth calling out:

- **Pausing stops future noise without touching history.** Upcoming actions are removed; everything already completed or missed stays exactly as it was.
 - **Deleting archives rather than destroys.** Progress you made does not disappear because you tidied up.
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4. The plan/log split

This is the decision the whole app rests on, and the reason it is not a CRUD app.

	Role	Example
Action	the rule	"Practise Spanish, 25 min, Mon–Fri, 07:30"
ActionOccurrence	one dated instance	"Wed 13 Aug, completed at 07:52"

A recurring action has no single status. "Gym on Mon/Wed/Fri" cannot answer "*did I miss Monday?*" if it is one row. Splitting the rule from the log makes it a lookup:

Feature	Becomes
Today's actions	one indexed query on today's date
Missed detection	a job flips <code>upcoming</code> → <code>missed</code> when the user's day ends
History	occurrences are never deleted, only status-changed
Calendar	occurrences already carry a date
Weekly progress	count planned vs completed in the week
Consistency streak	walk occurrence days backwards
Reminders	every occurrence has a scheduled time — that <i>is</i> the queue

Each occurrence carries one of five states: **upcoming**, **in progress**, **completed**, **missed**, **skipped**. A *skip* is treated as a decision, a *miss* as a failure — they are counted differently, because counting them the same punishes honesty.

5. Home dashboard

One screen, one question. Served by a **single API call** so the app opens instantly rather than chaining five requests.

- Personalised greeting with what is left today
- Today's actions with one-tap completion
- **Yesterday's misses carried over for one day**, so catching up is easy

- 7-day consistency strip and current streak
 - Active goals with progress ring and status chip
 - Goals needing attention, with the reason spelled out
 - Next milestone, recently completed items, unread notification count
 - One-tap theme switch and notification bell in the header
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6. Progress tracking

Four levels, each shown where it is useful:

Level	Example
Goal	<i>Learn Spanish — 35% complete</i>
Milestone	<i>Vocabulary — 18 / 30 actions completed</i>
Weekly	<i>This week — 22 / 33 planned actions</i>
Long-term	8-week trend, minutes invested, split by category

Visualisations are used only where they add understanding: a **ring** for goal progress, a **7-dot strip** for the week, a **thin bar** for milestones. Charts are confined to the Progress screen so the rest of the app stays calm.

7. Progress status — the part that is not a percentage

Every active goal is graded **Ahead** / **On track** / **Needs attention** / **Behind** / **Completed** from four independent signals:

```
timeRatio = how much of the goal's window has elapsed
workRatio = completed occurrences / all planned occurrences
adherence = last 14 days: completed / (completed + missed) [skips excluded]
delta     = workRatio - timeRatio
```

Condition (first match wins)	Status
<code>workRatio >= 1</code>	Completed
goal younger than 7 days	On track — grace period, new goals are never shamed
<code>delta >= +0.10</code>	Ahead
<code>delta >= -0.05</code> and <code>adherence >= 0.60</code>	On track
<code>delta >= -0.20</code> or <code>adherence >= 0.40</code>	Needs attention
otherwise	Behind

Every status ships with a plain-language reason the UI shows verbatim:

"You've completed 4 of 6 planned sessions this week." "Only 2 of your planned actions are done with 60 days left. Consider easing the routine rather than dropping the goal."

A coloured chip on its own explains nothing. The sentence is the feature.

8. Consistency, without gamification

Current streak	20 days
This week	22 / 33 planned
Last 30 days	71%

A day counts **only if something was planned for it**. Days with nothing planned are neutral — they neither break nor extend a streak. A rest day you scheduled yourself is not a failure.

There are no points, badges, levels or leaderboards. The number is there to be true, not to be maximised.

9. Calendar & schedule

A month grid showing planned, completed and missed actions plus upcoming milestones. Tap a day to see its actions, and complete, skip or **reschedule** from there.

It deliberately does not try to be Google Calendar. It exists only to make your goal-related schedule legible.

10. Weekly reflection

A dedicated screen, generated automatically each week:

- Goals worked on, actions completed and missed, completion rate
- Strongest area and the area needing attention
- Time invested, and next week's upcoming priorities
- Three optional prompts: **what went well** · **what was difficult** · **what to improve**

Reflections are saved per week and kept as history, so you can read back what you said three weeks ago and see whether you did anything about it.

11. Notifications

Everything is scheduled by the backend from the user's saved preferences. Nothing is hardcoded in the UI.

Type	Channel	When
Action reminder	push	N minutes before, N is user-set
Daily summary	push	at the user's chosen morning time
Weekly digest	push + email	user-chosen weekday and time
Milestone reached	push + email	on completion
Login greeting	push	on sign-in — see below
Verify email, password reset	email	transactional

The login greeting

The same "Welcome back" every time makes an app feel like it is not paying attention, so the message is chosen from how long you were away:

Gap	Message
First ever sign-in	<i>Welcome to GoalFlow, Priya</i> — set your first goal and we will turn it into something you can actually do each day
Same day	<i>Hey again, Sam</i> — 4 things still open today
1–3 days	<i>Welcome back, Sam</i> — 3 actions planned today, starting with Learn Spanish
4–14 days	<i>Good to see you, Sam</i> — it has been 9 days, pick one and you are moving again
15+ days	<i>Welcome back, Sam</i> — 40 days away, no catching up needed, just start from today

Copy for a long absence is written to remove guilt rather than add it. Shaming someone on the day they came back is how you lose them again.

Preferences

Push and email toggles, reminder lead time, daily summary time, weekly digest day and time, milestone alerts, and a **quiet hours** window that correctly handles overnight ranges. Reminders and digests respect quiet hours; a greeting you triggered by signing in does not, because you are demonstrably awake.

12. Appearance

Light / Dark / System, switchable from the home header in one tap or picked directly in Settings. The choice is stored on the device, survives sign-out, and applies instantly.

13. Authentication

Sign up, email verification by 6-digit code, login, logout, password reset by emailed link, password change, and optional Google sign-in.

Concern	Approach
Passwords	bcrypt, cost 12, never returned or logged
Access token	JWT, 15 minutes
Refresh token	Random, stored only as a SHA-256 hash, rotated on every use
Password change	Signs out every device
Account enumeration	Identical responses for known and unknown emails
Brute force	Rate limiting on every credential route
Authorization	Ownership checked per resource

14. Personalisation summary

Everything the brief asks to be customisable is, and is used downstream:

preferred days · preferred time of day · exact start time · frequency (daily / specific days / N per week / one-off) · goal priority · action difficulty · estimated duration · weekly target · reminder lead time · progress style · quiet hours · personal constraints · theme

Two users with the same goal get genuinely different schedules, reminders and dashboards.

15. Screens

Group	Screens
Auth	Splash · Register · Verify email · Login · Forgot password
Setup	Onboarding (5 steps)
Core	Home · Today · Goals · Goal detail · Calendar · Progress · Weekly reflection
Editors	Create/edit goal · Create milestone · Create action
Account	Profile · Settings · Notification preferences · Notifications feed

16. End-to-end journey

Open app → Create account → Verify email → Onboarding

- Goal created with routine and starter breakdown
- Personalised dashboard
- Complete actions, watch progress and status update live
- Calendar and Progress views
- Weekly reflection
- Adjust the goal or routine, schedule rebuilds

Every step runs against the deployed backend with no placeholder screens.